

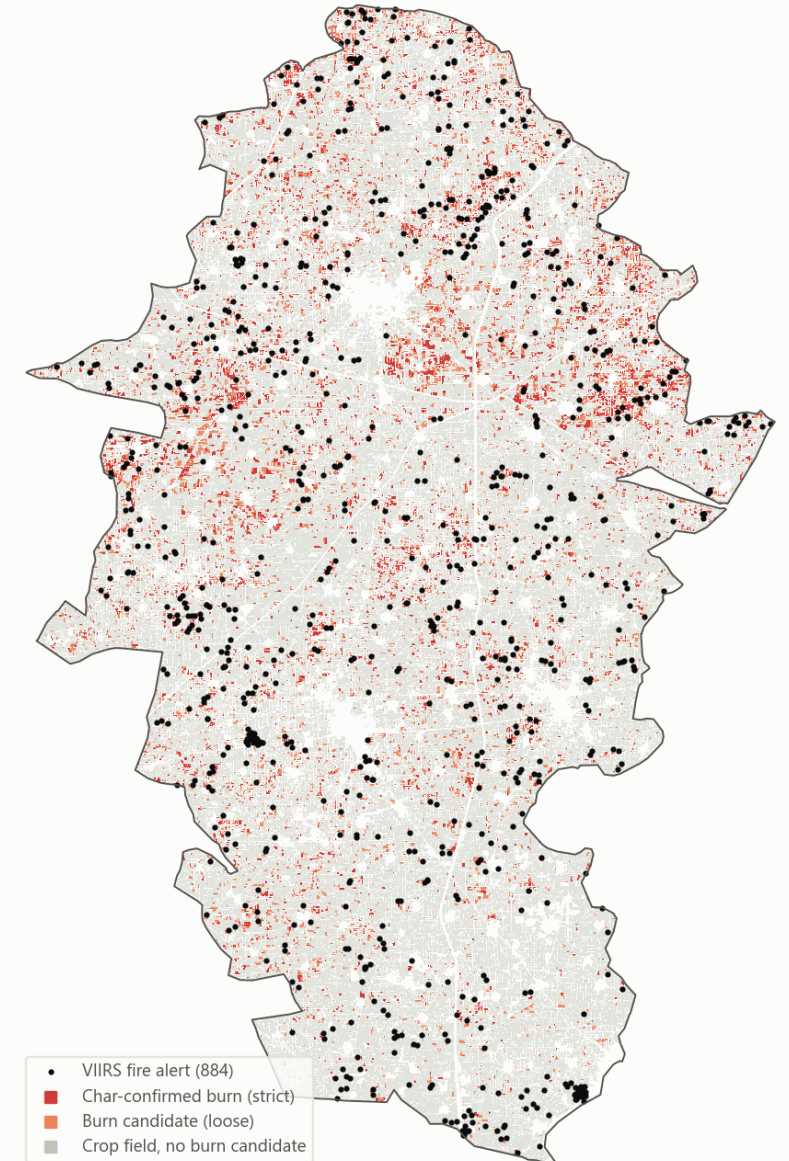
Parali Se Paisa

The burns the fire satellites stopped seeing,
and the balers that can reach them first

Satellite AI that turns paddy straw into verified farmer income
Greenovators Hackathon 2026 · Waste to Wealth · Net Zero AI
Architecture

Real data: Sentinel-2, VIIRS, MODIS and ERA5 for Sri Muktsar Sahib, Punjab, 2023–
2026

165,025 crop fields · 74 satellite scenes · 12.6 million field observations



The official indicator is going blind

18.8 Mt

paddy straw expected in Punjab in 2026,
most of it with a 2–3 week window before
wheat sowing

–92%

active-fire counts in Punjab since the 2021
peak (MODIS/VIIRS)

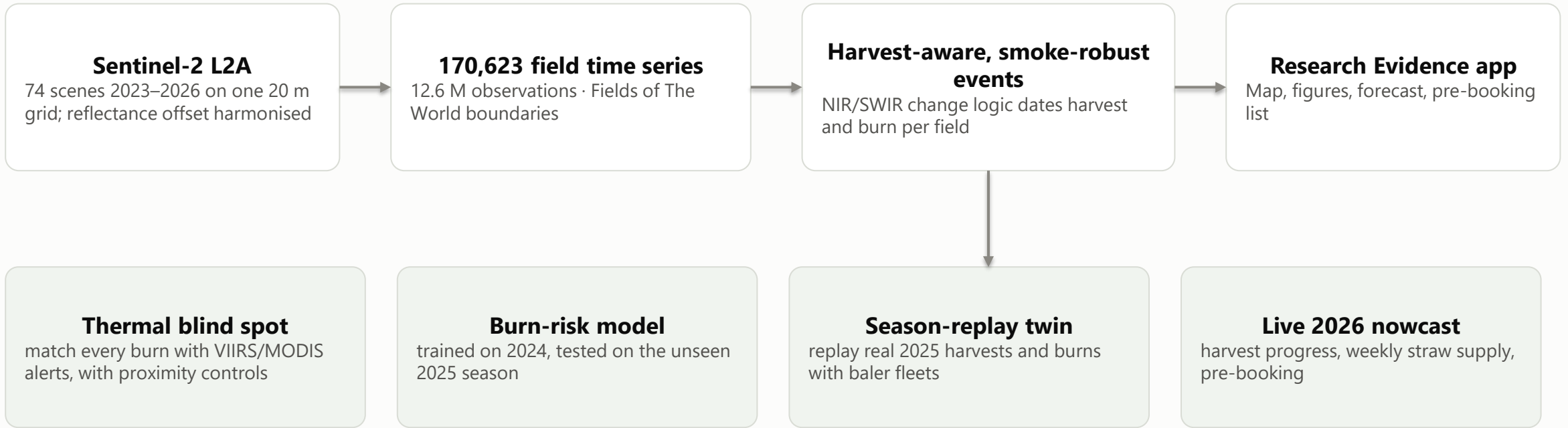
>90%

of large farm fires in 2024–25 lit after 3 PM,
after polar satellites pass (3% in 2021)

Burnt area fell only 25–35% while fire counts fell more than 95%. Money, machines and monitoring cannot target fields nobody can see.

Question: can free satellite data see every field's burn, and send balers before the fire?

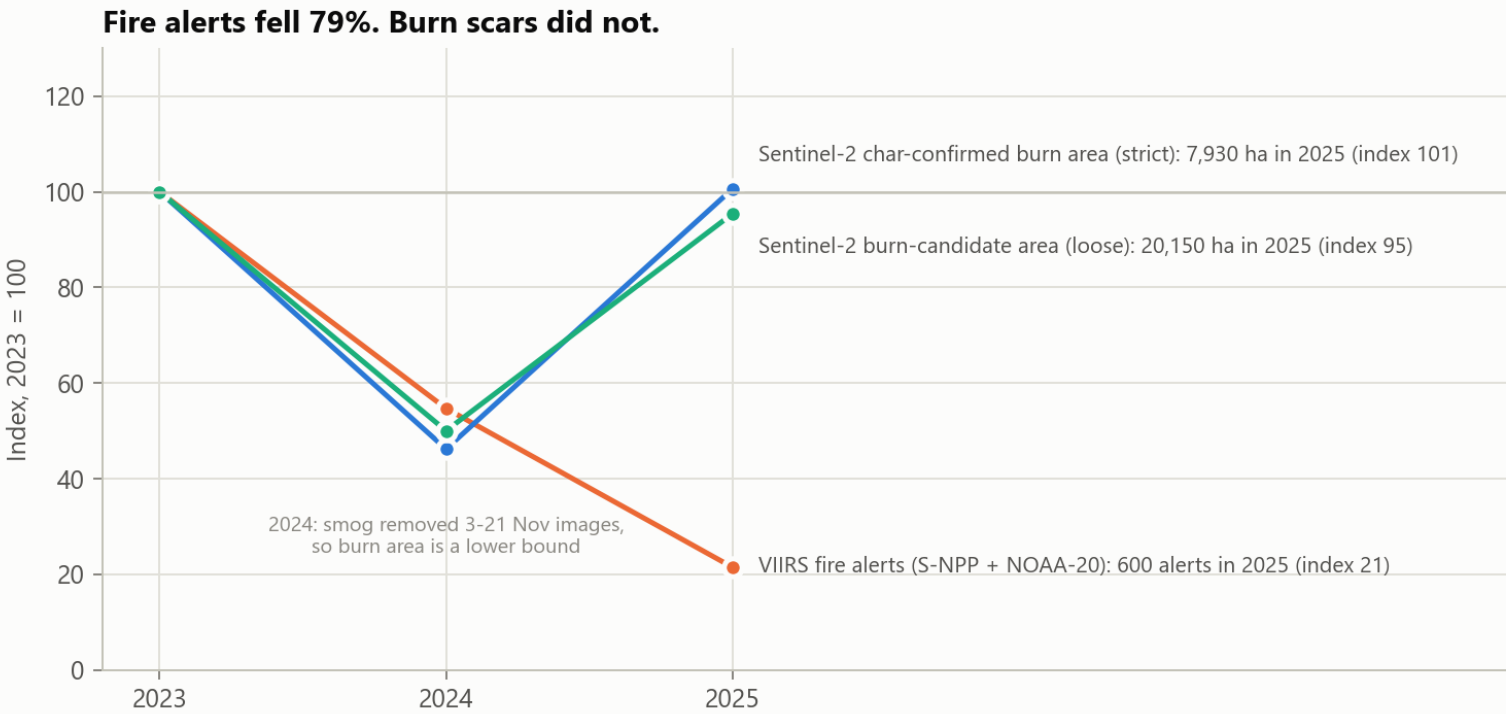
Every field, every 2–5 days, from free satellite data



Inputs: Sentinel-2 (ESA), VIIRS S-NPP / NOAA-20 / NOAA-21 and MODIS (NASA), MCD64A1, ERA5. All free and global, so it scales to any district.

FINDING 1

Fire alerts fell 79%. Burn scars did not.



Sri Muktsar Sahib, Oct-Dec. Sentinel-2 L2A at a harmonised 5-day revisit; VIIRS 375 m active fires (NASA FIRMS 2023-24, UMD archive 2025), vegetation-fire type, inside the district. Strict = char-confirmed tier; loose = all burn candidates.

2,792 → 600 VIIRS fire alerts in the district

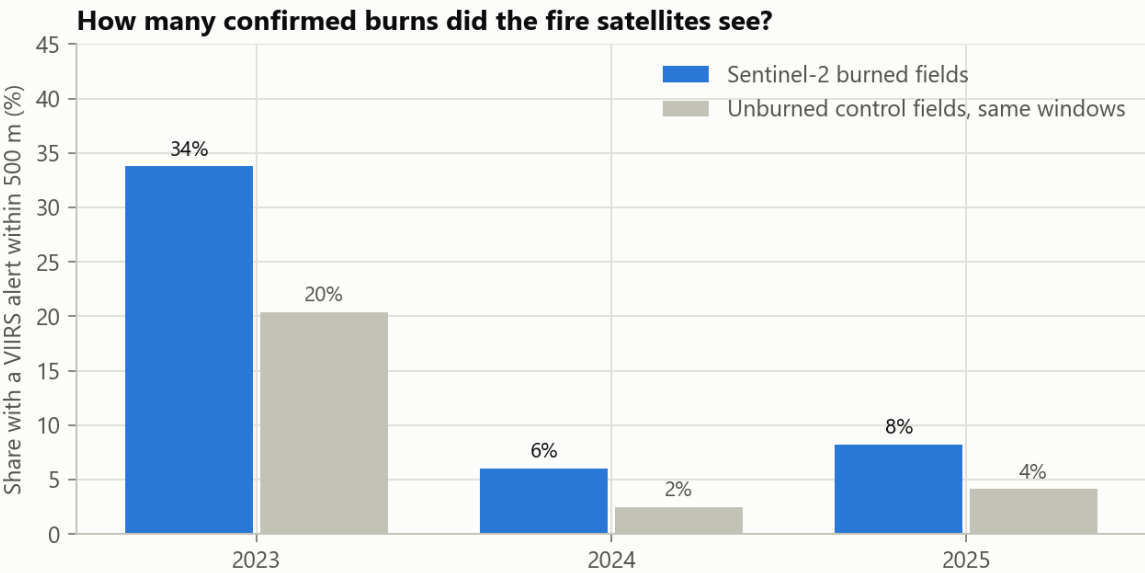
7,883 → 7,930 ha char-confirmed burn area (index 101)

2.3 → 11.88 burned fields per fire alert

Same revisit, same thresholds, same fields

FINDING 2

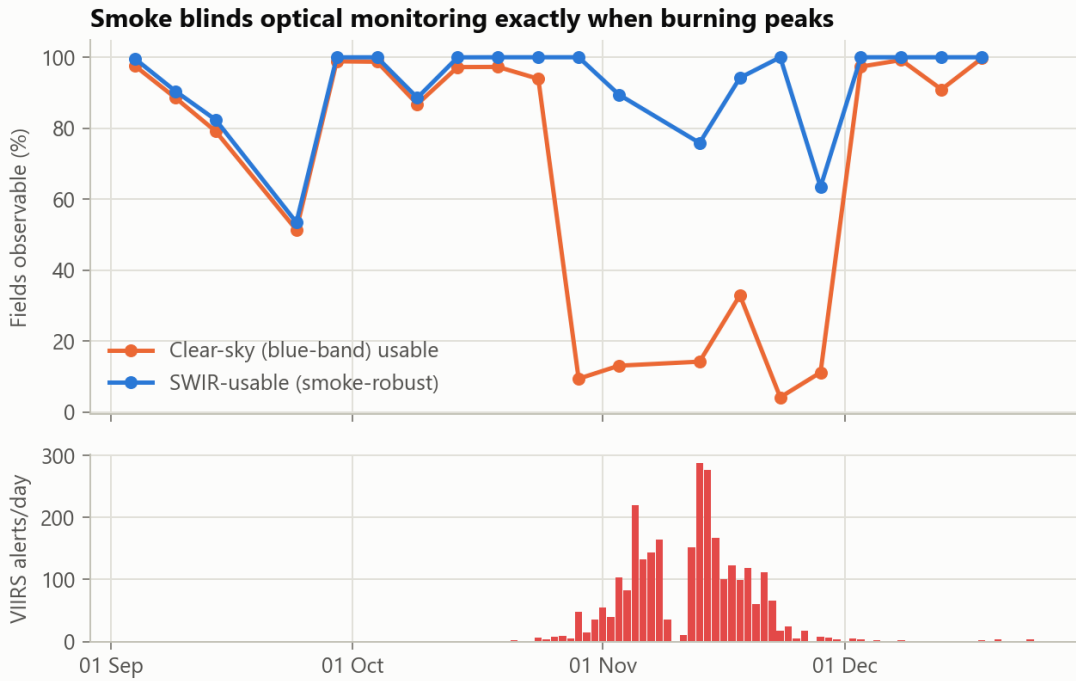
9 in 10 confirmed burns in 2025 raised no fire alert



Burned fields observed within <=5 days by Sentinel-2 (strict char tier).
Control: unburned harvested fields given the same observation windows (proximity baseline).

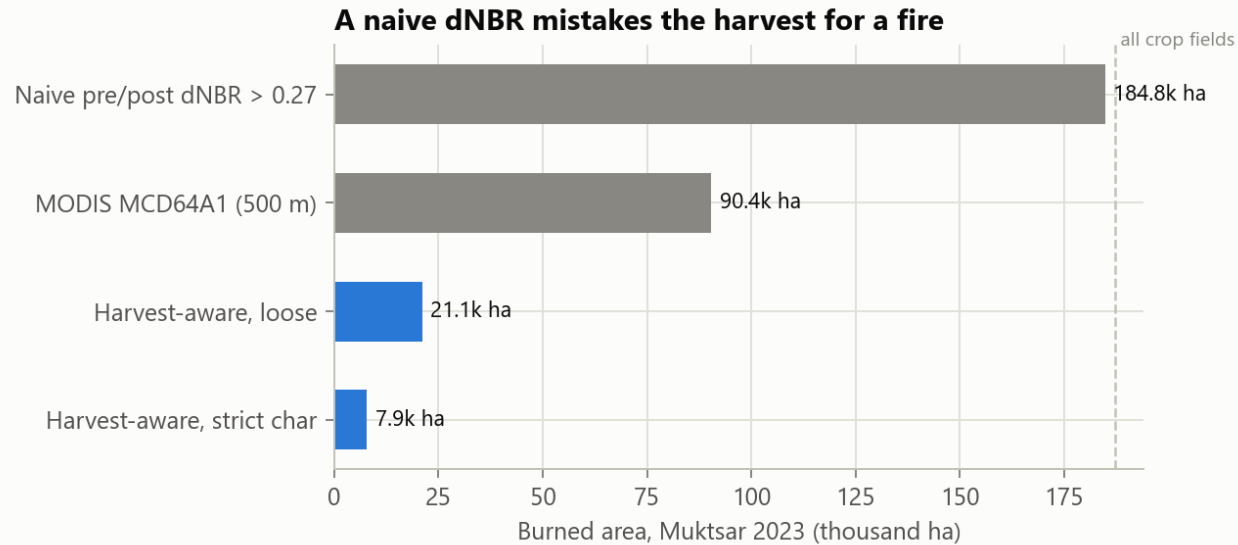
34% → 8% confirmed burns with any VIIRS alert (control 4%)

Independent of iFOREST: field-level evidence of after-overpass burning



2023 season, Muktsar crop fields. Sen2Cor cloud masks do not flag smoke; blue reflectance >= 0.10 marks hazy observations. SWIR (2.2 um) is little affected by smoke, so harvest and char logic uses NIR/SWIR only.

Harvest-aware detection, independently checked



Paddy harvest alone moves NBR from ~0.67 to ~0.1, so pre/post differencing flags almost every field.
Harvest-aware logic only scores post-harvest char. MCD64A1 area = burned 463 m pixels x nominal pixel area.

Smoke-robust: NIR/SWIR logic; Sen2Cor misses smoke haze

Wet-soil rejection: irrigated soil keeps a positive NBR, char does not

Thermal recall: 67% / 53% / 64% of VIIRS alerts have a Sentinel-2 burn within 500 m (2023/24/25)

Controls and sensitivity: proximity controls; radius 375–1,000 m; harmonised revisit

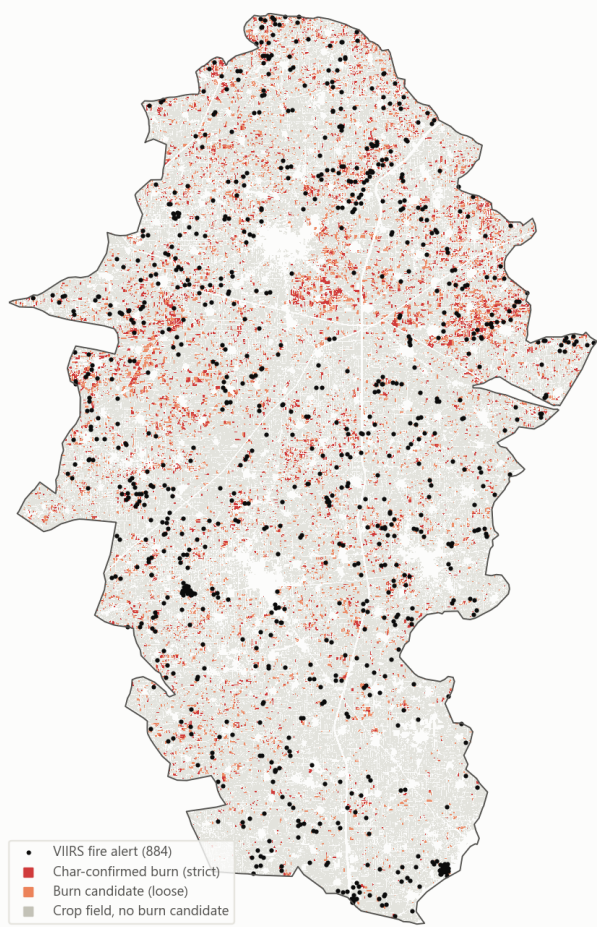
Human validation: stratified, blind before/after labelling tool → precision, recall, area with 95% CI

Bug found and fixed: missing –1000 DN reflectance offset inflated reflectance by 0.10

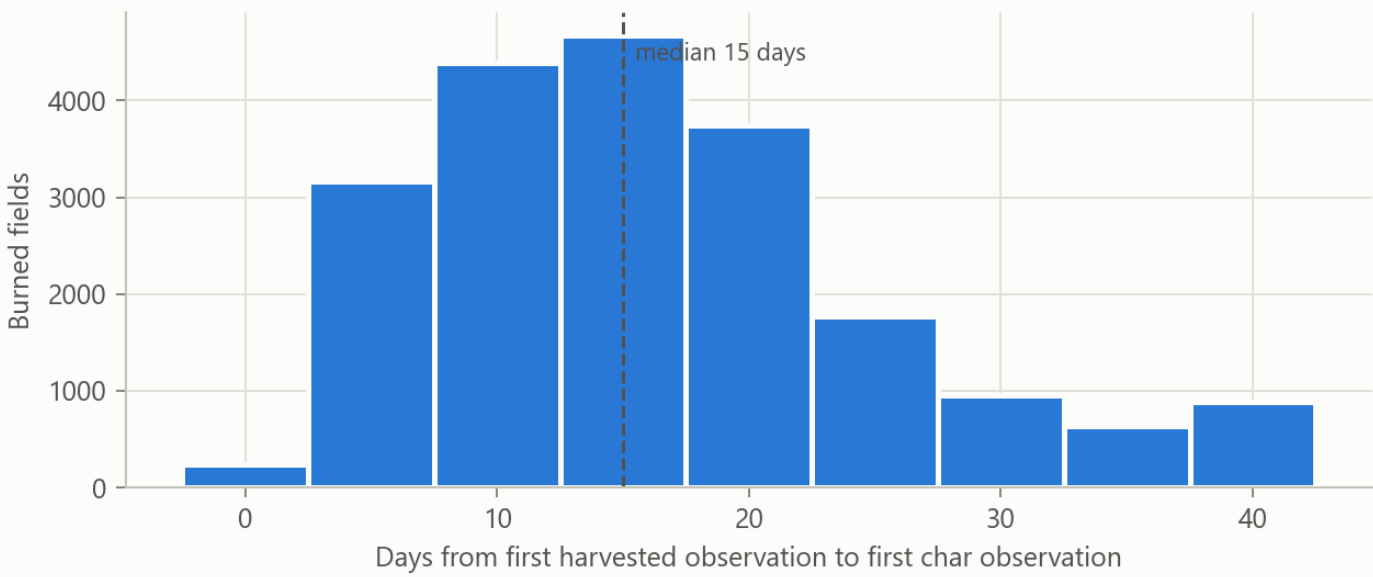
FINDING 3

Burning clusters, persists, and leaves about a two-week window

Muktsar 2025: burn scars mapped field by field



The intervention window: about two weeks between harvest and fire



2025, burn-candidate fields (loose tier). Event dates are interval-censored by the 2-5 day revisit.

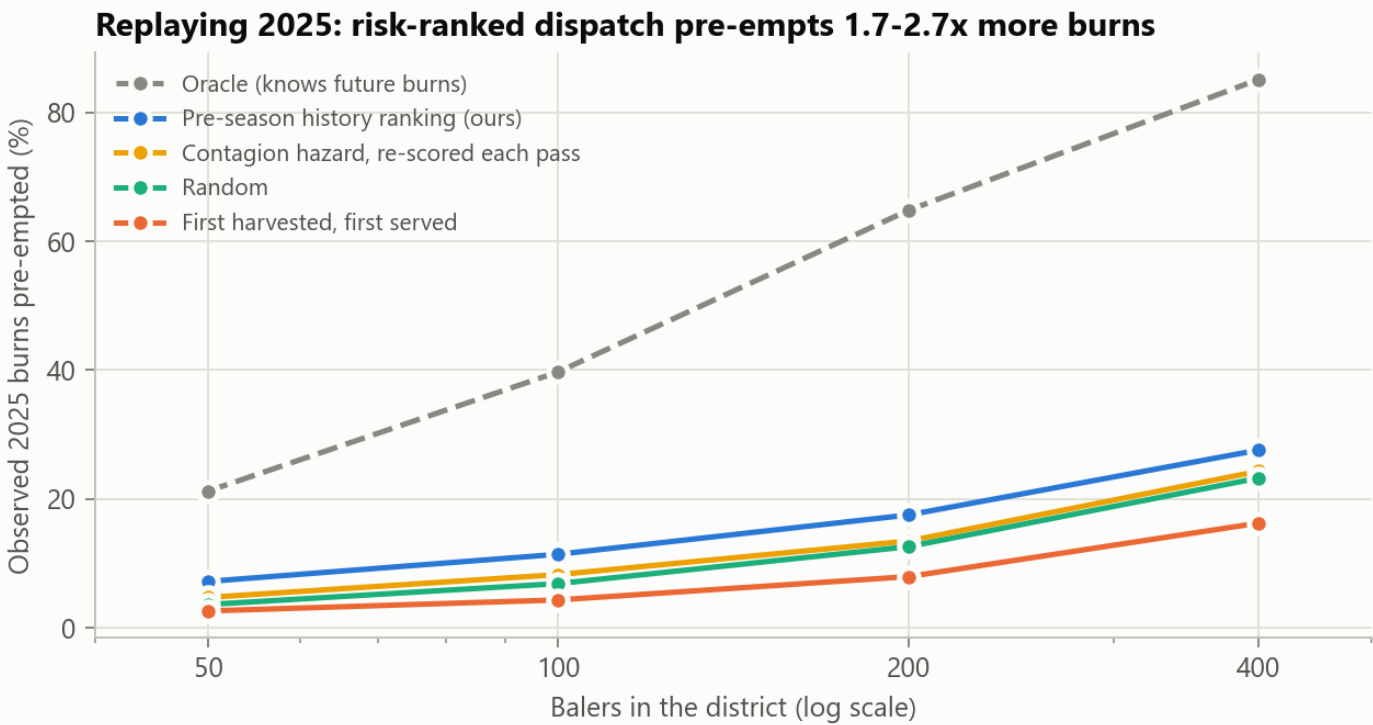
15 days median harvest → fire window (2025)

30% vs 12% burn again if burned last year vs not

4.6% of fields burned in 2+ seasons and did 34% of all burning (2023–25)

Sentinel-2 L2A harvest-aware burn detection on 165,025 crop fields (Fields of The World boundaries).
Black dots: VIIRS 375 m active-fire alerts (S-NPP, NOAA-20, NOAA-21), Sep-Dec 2025.

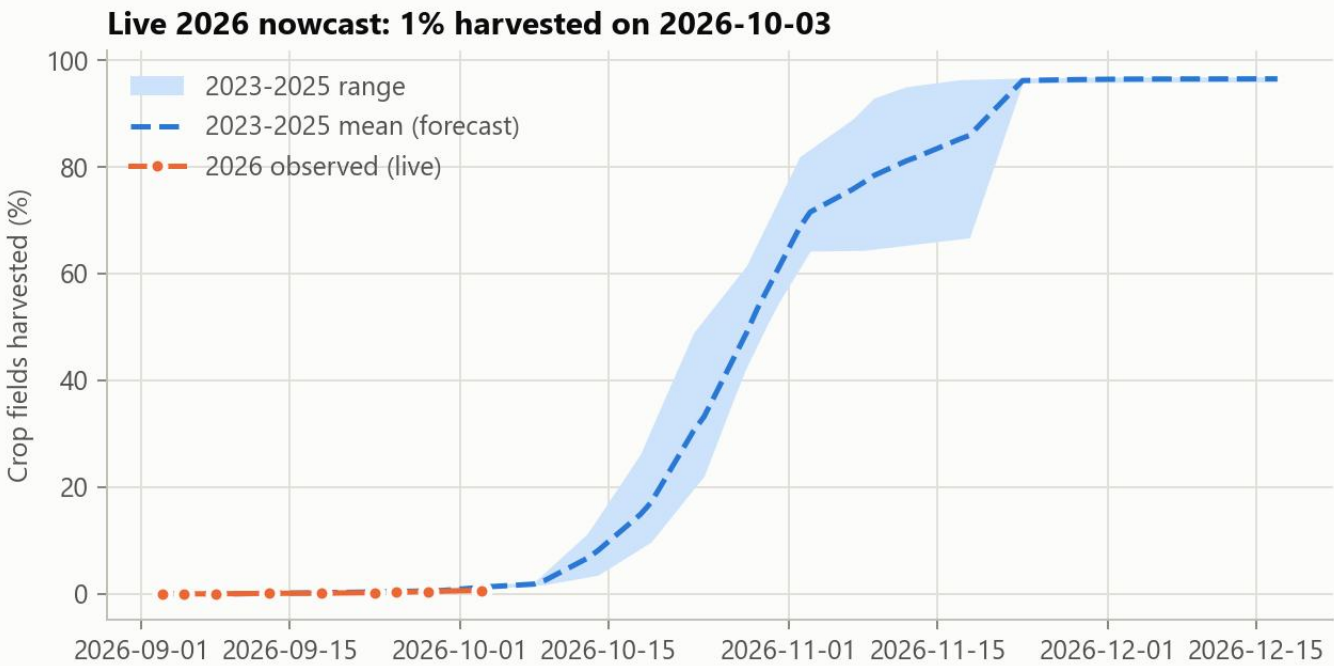
Risk-ranked dispatch pre-empts 2.6× more burns



Season-replay digital twin on real 2025 Sentinel-2 harvest/burn events; 4 ha/baler/day, ERA5 rain days skipped, 20-day pickup window. Assumes a baled field is not burned; policy comparison, not an impact forecast.

11.4% vs 4.3% burns pre-empted, 100 balers
1,859 t CO₂e CH₄+N₂O avoided (90%: 328–4,849)
85 t PM2.5 and 4.4 t black carbon avoided
₹11.73 crore of straw baled by the fleet at ₹169/quintal
Top decile = 2.7× lift on the unseen 2025 season (pre-season model)

Live: Muktsar 2026, as of this week's satellite pass



Muktsar crop fields; harvest observed when field NBR falls below 0.30.
Forecast assumes a typical (2023-2025) harvest pace.

Week of	Expected harvest	Baleable straw
10-03	4,284 ha	14,136 t
10-10	17,605 ha	58,095 t
10-17	36,572 ha	120,687 t
10-24	50,835 ha	167,755 t
10-31	27,075 ha	89,349 t
11-07	12,534 ha	41,363 t

Pre-booking list: top decile = 15,864 fields, 18,399 ha
District straw ≈ 11.5 lakh t (published estimate: 12.1 lakh t)

One data backbone for officials, balers, buyers and farmers



PARALI
SE PAISA

FIELD INTELLIGENCE

REAL DATA / WEAK PROXIES

WORKSPACE

Overview

Command Map

Fields

Dispatch

Verification

Certificates

Research Evidence

OPERATIONS

Balers

Buyers

PORTALS

Farmer Portal

OF

official

OFFICIAL

API ok · REAL DATA / WEAK PROXIES

Research Evidence

REAL DATA / WEAK PROXIES

Read-only real-data mode displays acquired Sentinel-2 and ERA5 observations with research field boundaries. Labels are weak proxies; no ground truth, verified buyer/baler registry, or operational decision model is available.

REAL SATELLITE EVIDENCE · SRI MUKTSAR SAHIB · 2023-2026

Research evidence

Dense Sentinel-2 time series for every field in the district, matched against VIIRS/MODIS fire alerts and replayed as a dispatch digital twin.

Research evidence from real satellite data. Burn tiers are rule-derived Sentinel-2 candidates, not ground truth and not enforcement evidence; they exist to prioritise straw pickup and payments.

VIIRS fire alerts, 2023 - 2025

-79%

2,792 - 600 alerts (S-NPP + NOAA-20)

Char-confirmed burn area, 2025 vs 2023

101

index · 7,930 ha strict, 20,150 ha loose

Confirmed burns with a VIIRS alert, 2025

8%

proximity control 4% · ≤5-day windows

Intervention window

15 days

median, first harvested -> first char observation (2025)

Burns pre-empted with 100 balers

11%

risk-ranked vs 4.3% first-come-first-served

2026 harvest, 3 Oct 2026

0.7%

same date: 2023 1.1% · 2024 0.9% · 2025 1.6%

Field-level burn-scar and risk map

2023 burn scars

2024 burn scars

2025 burn scars

2026 pre-season risk

☐ VIIRS fire alerts (884)

DEMO / SYNTHETIC

Satellite and provider status is unverified. Rule outputs are candidates; routes use a straight-line planning proxy.

FARMER PORTAL

ਤੁਹਾਡੇ ਖੇਤ ਦੀ ਜਾਣਕਾਰੀ

A simple view of your field, straw estimate, and pickup progress.

Language ਪੰਜਾਬੀ

ਤੁਹਾਡਾ ਦਰਜ ਖੇਤ

SYNTHETIC-ACTION-FIELD-01

SYNTHETIC TEST LOCALITY · 2 ha

Harvested candidate

ਅਨੁਮਾਨਿਤ ਪਰਾਲੀ

12 t

Preliminary scenario estimate

ਮੌਜੂਦਾ ਸਥਿਤੀ ਦਾ ਅਨੁਮਾਨ

Harvested candidate

Based on stored candidate signals.

Certificate record

Prototype Verification Certificate

REAL-2025C-PROXIMAL-LABEL-4488-015407C40C88

SYNTHETIC-ACTION-FIELD-01

1 Sept 2026

1 Oct 2026

2

No burn verified

FastAPI + React app; 197 backend tests plus new research tests; real-data mode is read-only; field-level burn data restricted to officials.

10

Waste to wealth at district scale, with no hardware

₹194 cr

value of Muktsar's ≈ 11.5 lakh t of straw at ₹169/quintal

741 t

CO₂e (CH₄ + N₂O) avoided per 1,000 ha not burned, plus 34 t PM_{2.5}

₹0

hardware: free Sentinel-2 / VIIRS / ERA5 and open-source tools

Who pays: state CRM schemes for verified no-burn incentives (Haryana pays ₹1,200/acre after inspection); buyers pay per traceable tonne for ESG supply chains

Scale: same code for any district; Punjab has 23 and Haryana 22, using the same satellite tiles and fire archives

SDGs: 12 responsible production · 13 climate action · 11 clean air in cities · 9 innovation

What we have not shown yet, and how we will

Precision: blind stratified labels in progress (AI-assisted check: about 7–8 of 10 strict detections show char)

Lower bounds: small or quickly tilled burns between revisits are missed

Crop type: the cotton-growing south needs a Sentinel-1 paddy mask

Replay assumptions: baling prevents burning; no travel time; no farmer acceptance yet

Week 1: finish labels → publish precision, recall and area with 95% CI

Month 1: Sentinel-1 radar for smog gaps and a paddy mask

Season 2026: pilot in one block with a baler cooperative and a straw buyer

Policy: propose burn-scar area, not fire counts, as the official indicator

Measure burns, not alerts. Pay for straw, not fines.

Fire alerts –79% · burn scars unchanged · 92% of confirmed burns unseen by fire satellites (2025)

2.6× more burns pre-empted with the same 100 balers · live 2026 nowcast for Muktsar

Parali Se Paisa · Greenovators Hackathon 2026 · Amity University Noida

Reproducible: [scripts/run_research.py](#) · [docs/RESEARCH_METHODS.md](#) · [reports/research/RESEARCH_SUMMARY.md](#)